university of groningen

Potential-Based Reward Shaping for Planar Pushing

Single-Agent Control vs. Asymmetric Self-Play

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1. Introduction & Research Questions

Outline

1. **Introduction & Research Questions**
2. **Background & Related Work**
   reward shaping · curriculum & self-play · push mechanics & SE(2)
3. **Problem Definition & MDP**
4. **Implementation**
   six models: PPO-Baseline, PPO-PBRS, PPO-Curriculum, ASP-dPose, TASP-dPose, TASP-dPose-BP
5. **Evaluation**
   30 held-out scenes · 20 trials · Isaac vs gym
6. **Results**
7. **Discussion**
8. **Conclusion & Takeaways**

Research Questions

Three Research Questions

1. **RQ1:** Can potential-based reward shaping improve the efficiency of hand-tuned dense rewards for goal reaching tasks in a robotic enviroment?
2. **RQ2:** Given a well-shaped PBRS reward function does adding a curriculum or asymmetric self-play improve final task success over a direct single-agent approach?
3. **RQ3:** Can a sufficiently informative reward (PBRS) *substitute* for an explicit curriculum in contact-rich pushing?

[Ng, Harada & Russell 1999] **[Grześ 2017]** [Devlin & Kudenko 2012]

2. Background & Related Work

Related Work – Reward Shaping (PBRS)

Potential-based reward shaping :

$$\mathcal{R}'(s,a,s') = \mathcal{R}(s,a,s') + \underbrace{\gamma\Phi(s') - \Phi(s)}_{F}$$

- **Policy invariance:** $Q'(s,a) = Q(s,a) - \Phi(s)$ – the optimal policy is unchanged for *any* state potential Φ.
- **Episodic validity** : $\gamma_{\text{shaping}}=1$ is safe – $\sum F = \Phi(s_T) - \Phi(s_0)$ is bounded.
- **Dynamic / arbitrary potentials** – basis for our exponential Φ.

Why it matters here: lets us design a dense, well-behaved reward *without* changing the task's optimum – the foundation of every model in this thesis.

[Ng et al. 1999] **[Grześ 2017]** [Devlin & Kudenko 2012] [Harutyunyan et al. 2015]

Related Work – Curriculum & Asymmetric Self-Play

Forced curricula

- **Surveys**
- **Reverse / goal** : start near the goal, expand outward
- **Precision** : gradually tighten the success tolerance

Our use: a forced 3-phase pos→rot curriculum (PPO-Curriculum)

Automatic (self-play)

- **ASP** : Alice/Bob adversarial game + ABC
- **Time-based Alice** : $R_A \propto t_B - t_A$, self-regulating
- **Regret-based**

Our use: ASP-dPose (outcome) and TASP-dPose (time-based)

[Narvekar et al. 2020] [Portelas et al. 2020] **[Sukhbaatar et al. 2018]** [Plappert et al. 2021]

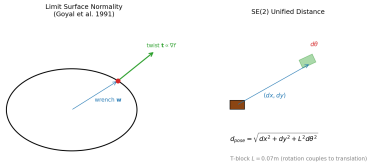
Related Work – Geometry & Mechanics of Planar Pushing

Push mechanics

- **Quasi-static pushing**: motion set by the friction limit surface.
- **Limit-surface normality**: *every* push couples translation and rotation.

SE(2) geometry

- **Unified distance**: a single scalar that combines position and orientation error, weighted by the object's physical size *L*
- **Equivariant manipulation**: exploiting SE(2)/SO(2) symmetry.



3. Problem Definition & MDP

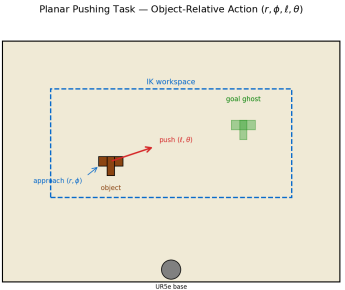
Planar-Pushing MDP – State, Action & Transition

State $\mathbf{s} \in \mathbb{R}^{28}$: EE pose (6) | object pose (14) | goal pose (6) | errors (2)

Action $\mathbf{a} \in \mathbb{R}^4$ (object-relative): (r, ϕ, ℓ, θ) – 21 bins/dim, 194 481 discrete actions

Transition : $\mathbf{t} \propto \nabla f(\mathbf{w})$ – limit-surface normality couples **every** push with rotation

Success: pos $<$ 5 cm **and** rot $<$ 0.2 rad.
Validation: 30 scenes $\times$ 20 trials $\times$ up to 30 pushes



[Mason 1986] [Lynch & Mason 1996] [Goyal et al. 1991]

Why the Fractional Reward Failed

Fractional improvement formula:

$$R = \underbrace{\alpha \cdot \frac{d_{\text{prev}} - d_{\text{now}}}{d_{\text{prev}}}}_{\text{improvement}} - \underbrace{\beta_{\text{pos}} \cdot d_{\text{now}}}_{\text{pos penalty}} - \underbrace{\beta_{\text{rot}} \cdot y_{\text{now}}}_{\text{rot penalty}}, \quad \alpha{=}3.0,\ \beta_{\text{pos}}{=}0.5,\ \beta_{\text{rot}}{=}0.25$$

Worked example – pushing toward a goal 30 cm away, 1 cm improvement:

$$R = 3.0 \cdot 0.01/0.30 - 0.5 \cdot 0.29 - 0.25 \cdot 0.5 = \mathbf{-0.17}$$

Problem 1 – Negative reward for progress: the penalty terms dominate the improvement.

Problem 2 – Near-zero variance: 99% of pushes produce $R \in [-0.5, +0.1]$.

[Ng et al. 1999] [Grzes & Kudenko 2009] [Schulman et al. 2017]

How PBRS Fixes the Reward

PBRS potential: $\Phi(s) = w \cdot e^{-k \cdot d^2}, \quad k{=}30,\ w{=}10$

Same push, same 1 cm improvement:

$$R = w \cdot \left[e^{-k \cdot d_{\text{now}}^2} - e^{-k \cdot d_{\text{prev}}^2} \right] = 10 \cdot \left[e^{-30 \cdot 0.29^2} - e^{-30 \cdot 0.30^2} \right] = +\mathbf{0.133}$$

Why this matters:

- **Correctly signed** – every push toward the goal gets positive reward, away gets negative
- **Bounded** – $\Phi(s) \in [0, w]$, so $|R| \leq w$ regardless of distance or physics glitches
- **Markovian** – depends only on current state s, not on history or d_{prev}
- **Policy-invariant** – the optimal policy under PBRS is identical to the original task

[Ng et al. 1999] [Grześ 2017] [Harutyunyan et al. 2015]

4. Implementation

Baseline – Fractional Reward at Scale

Same fractional reward, same PPO+LSTM, same 2,048 envs, same 100K iter budget:

Model	Scene SR	Pos-only	Pos+rot
PPO single-agent	**80.0%**	100%	70%
ASP (Alice+Bob)	**0.0%**	0%	0%

Same PPO+LSTM, same push primitive, same training budget – only the architecture differs. **The adversarial goal distribution, not the reward formula, is the bottleneck.** *(See Appendix K for training budget comparison.)*

[Sukhbaatar et al. 2018] [Plappert et al. 2021]

All Models at a Glance

Model	Idea	Reward / curriculum
PPO-Baseline	single agent, original reward	fractional improvement (non-PBRS)
PPO-PBRS	single agent, no curriculum	two-potential PBRS (pos, rot)
PPO-Curriculum	single agent + staged curriculum	PBRS + forced pos→rot
ASP-dPose	self-play (Alice/Bob)	SE(2) PBRS, outcome Alice ($+5/-1$)
TASP-dPose	self-play (Alice/Bob)	SE(2) PBRS, time-based Alice
TASP-dPose-BP	self-play + Bob penalty	SE(2) PBRS, full Sukhbaatar symmetric

All six models share the same PPO+LSTM backbone and object-relative push primitive.

[Schulman et al. 2017] [Ng et al. 1999] **[Sukhbaatar et al. 2018]**

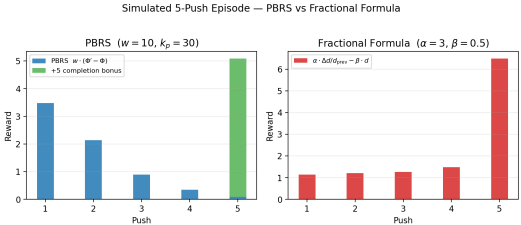
PPO-PBRS – Single-Agent (Recommended Baseline)

Architecture: Single PPO agent, LSTM(28→256) + MLP[512, 256] trunk, MultiCategorical head (4 dims × 21 bins). No curriculum, no Alice/Bob, no GoalEncoder.
Training: 528 envs, 15 pushes/rollout, ∼3000 iterations ∼1.6 days wall-clock (∼0.022 it/s).
Why it works:

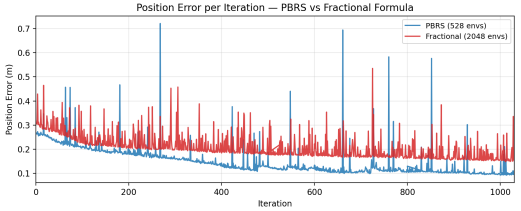
1. **Dense, bounded gradient** – PBRS fires on every push
2. **Fixed goal distribution** – stationary random goals; no adversarial shift
3. **Single network** – one agent, one optimizer; no multi-agent instability

[Ng et al. 1999] [Grześ 2017] [Schulman et al. 2017]

PBRS vs Fractional– Reward Signal Analysis



Per-push reward – Fractional vs PBRS: same 5-push episode. Fractional formula: negative or zero for 80% of pushes. PBRS: correctly-signed, bounded on every push.



Position error per iteration (TensorBoard): PBRS at 528 envs converges to lower PosErr than fractional formula at 2,048 envs – 4× more environment-efficient.

Dense signal on every push, bounded rewards, stable value – PBRS avoids the gradient starvation and critic instability of the old fractional formula.

[Ng et al. 1999] **[Grzes 2017]** [Sabulman et al. 2017]

Why PBRS for ASP? – Decoupling Advantage from Value

GAE advantage decomposes into two parts:

$$\hat{A}_t = \underbrace{\sum (\gamma\lambda)^l \left[\Phi(s_{t+l+1}) - \Phi(s_{t+l})\right]}_{\text{always correctly-signed}} + \underbrace{\sum (\gamma\lambda)^l \left[\gamma V(s_{t+l+1}) - V(s_{t+l})\right]}_{\text{biased when } V \text{ is stale}}$$

Φ-term: PBRS potential difference

$\Phi(s) = w \cdot e^{-k \cdot d^2}$ increases when object moves toward goal. Correct sign on every push regardless of $V(s)$ accuracy.

V-term: value function bootstrap

Under ASP, Alice shifts goal distribution $\rightarrow$ $V(s)$ perpetually stale. GAE bootstraps from wrong values $\rightarrow$ biased advantage.

Outcome: PBRS lifts ASP 0% $\rightarrow$ 16.7% via the Φ-term – but only partially, because the V-term bias remains.

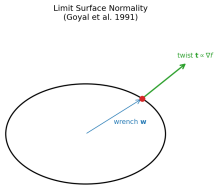
Single-agent: no distribution shift $\rightarrow$ V accurate $\rightarrow$ both align $\rightarrow$ 80%.

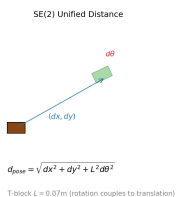
SE(2) Unified Distance – Self-Play Variants

SE(2) metric: $d_{\text{pose}} = \sqrt{dx^2 + dy^2 + L^2 d\theta^2}$

- Single scalar combining (dx, dy) and $d\theta$; L = char. length (T-block 0.07 m)
- **Single-potential PBRS:** $\Phi(s) = w\, e^{-k\, d^2_{\text{pose}}}$, $k = 30,\ w = 10$
- Eliminates competing pos/rot gradients – aligns reward with the coupled physics

Result: SE(2) does *not* rescue self-play – ASP-dPose reaches only 6.7% validation SR vs single-agent 80%. The self-play curriculum is the root cause, not the metric.





[Park 1995] [Urain et al. 2023] [Lynch & Mason 1996]

PPO-Curriculum – Position→Rotation Curriculum

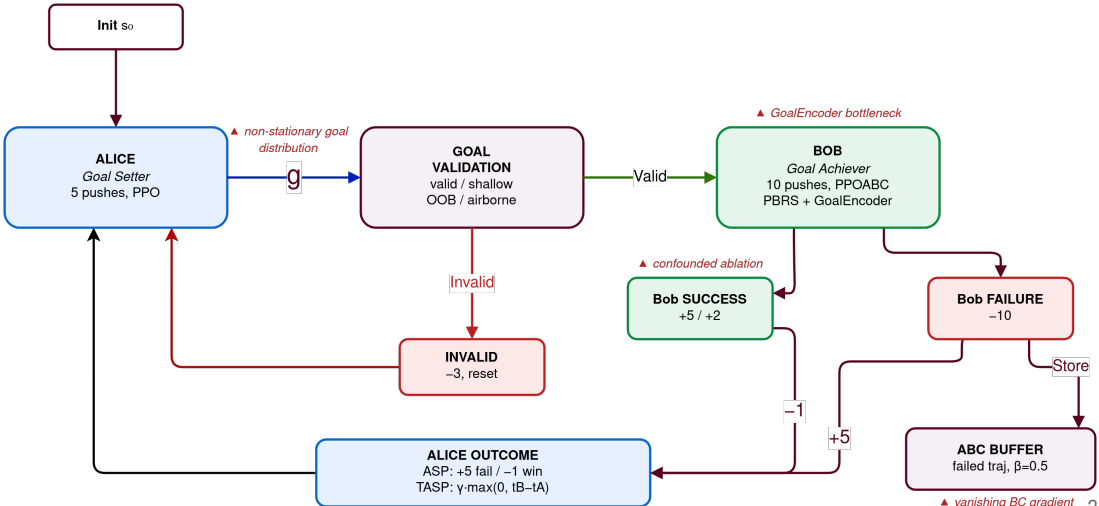
Design: Same PPO agent as PPO-PBRS + 3-phase curriculum:

1. $w_{\text{rot}} = 0$, position-only training
2. Triggered when episodic position-SR ≥ 0.5 for 20 iters: w_{rot} ramps $0 \rightarrow 10$ over 200 iters
3. Full PBRS (multi-objective)

Trigger: gates on episodic position-only SR $\geq$ 0.5 over a 20-iter lookback. Curriculum now activates and reaches 76.7% SR – only 3.3pp below PPO-PBRS. *(See Appendix H for original trigger bug details.)*

[Narvekar et al. 2020] [Portelas et al. 2020] [Florensa et al. 2017]

ASP Loop – Alice↔Bob Cycle



ASP-dPose vs TASP-dPose – Alice Reward

Both share the SE(2) d_{pose} Bob reward and T-block object; only Alice's reward differs.

Self-play variant	Alice reward	Valid SR
ASP-dPose (outcome-based)	$R_A = +5$ fail $/ -1$ win, $\beta_{\text{abc}}=0.5$	6.7%
TASP-dPose (time-based)	$R_A = \gamma_{sp} \cdot \max(0, t_B - t_A)$, no ABC	16.7%

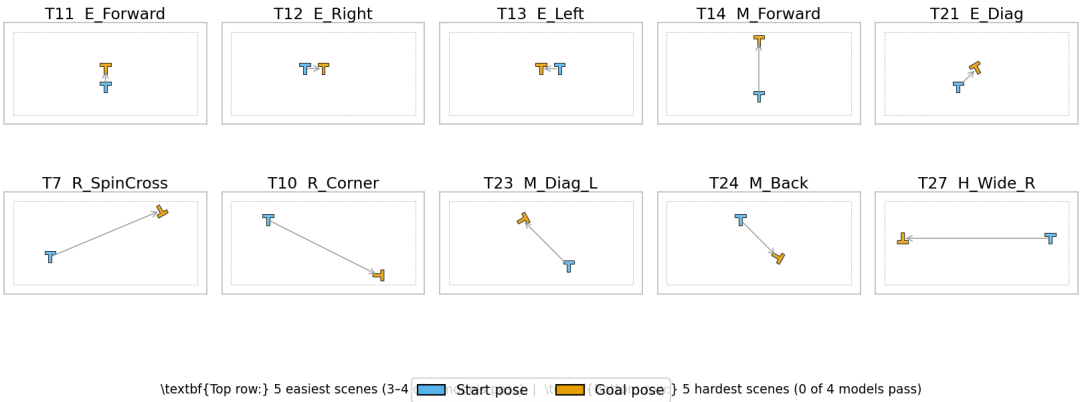
Findings:

- Time-based Alice (TASP-dPose) outperforms outcome-based (ASP-dPose) by $+10$pp
- TASP-dPose disables ABC – the gain comes from the time-based reward alone
- Best self-play variant (TASP-dPose, 16.7%) still **4.8**$\times$ below single-agent (PPO-PBRS, 80.0%)

[Plappert et al. 2021] **[Sukhbaatar et al. 2018]** [Dennis et al. 2020]

5. Evaluation

Validation Test Suite – Top 10 Held-Out Scenes



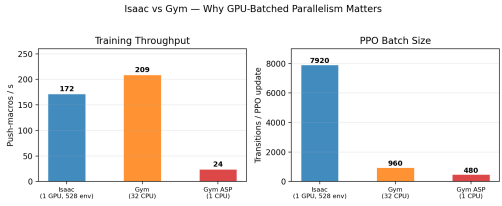
T-block **start pose** → **goal pose**. Every model is evaluated on an identical 30-scene suite (20 trials × up to 30 pushes per scene).

Isaac Lab – the Right Venue for Self-Play

GPU-batched parallelism – 528 environments stepped simultaneously on one RTX Pro 6000. Produces 7,920-sample PPO batches – 8× larger than gym-pusht (960).
Large batches stabilize PPO+LSTM gradients – essential for the non-stationary two-agent self-play objective.

Robotic fidelity – UR5e arm + 3D contact physics + cuRobo IK = the actual SE(2) task. gym-pusht is a 2D diagnostic only; it cannot test self-play under realistic contact dynamics.

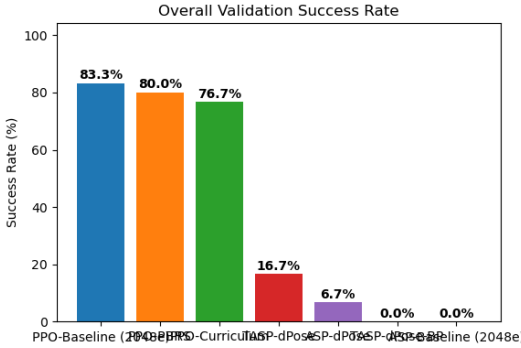
Fair budget – self-play and single-agent run at identical per-iteration wall-clock (∼0.022 it/s). Isaac gives self-play its **best shot**, matched to the winning single-agent –



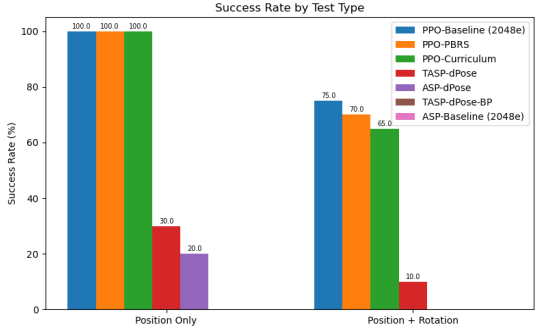
Throughput is comparable; batch size is the differentiator.

6. Results

Multi-Model Validation



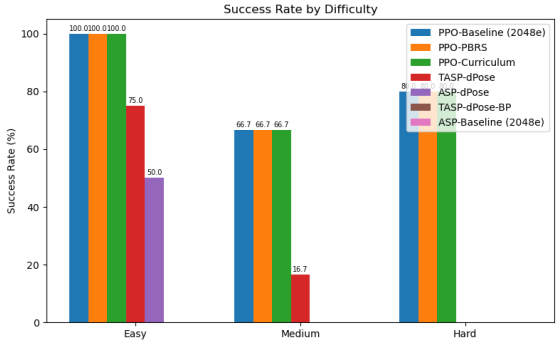
Overall SR & SE – 30 T-block scenes, best-of-20



SR by test type: pos-only (10 tests) vs pos+rot (20 tests)

PPO-PBRS vs PPO-Curriculum: 80.0% vs 76.7% – 3.3pp gap **not statistically significant** ($p > 0.05$ on 30 independent scenes). **Self-play (ASP-dPose, TASP-dPose):** 4.8–11.9× below

Comparison Summary – All Models with Error Bars



Key numbers (30 T-block scenes):

Model	SR	SE	95% CI
PPO-Baseline	80.0%	7.3%	65–95%
PPO-PBRS	80.0%	7.3%	65–95%
PPO-Curriculum	76.7%	7.7%	61–92%
TASP-dPose	16.7%	6.8%	3–30%
ASP-dPose	6.7%	4.6%	0–16%
TASP-dPose-BP	0.0%	–	–

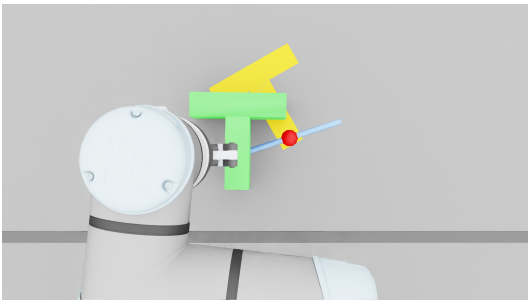
PPO-PBRS vs PPO-Curriculum gap: 3.3pp – **not statistically significant** ($p > 0.05$). Both solve the task similarly. Self-play variants are 4.8–11.9× below – this gap **is** significant.

Error bars = ±1 standard error across scenes. Base ASP and the GoalEncoder-ablation are excluded – neither has a

PPO-PBRS – Validation Results

30 held-out T-block scenes (20 trials each):

- **80.0% SR** (24/30) – pos-only **100%**, pos+rot **70%**
- PosErr **0.032 m**, RotErr **0.568 rad** (mean over all 30 scenes, incl. failures)
- Per-attempt (trial-level) SR **66%**; scene SR counts a scene solved if any of its 20 trials passes

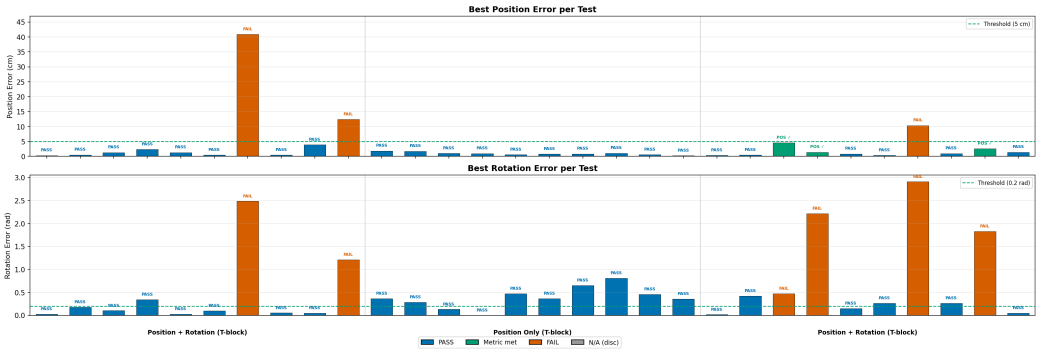


Overhead view – PPO-PBRS solving a pos+rot scene ▶

play clip

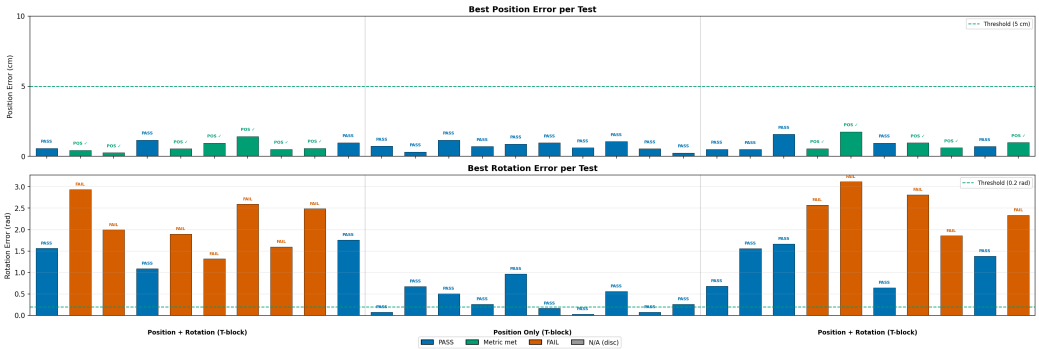
[Ng et al. 1999] [Grześ 2017] [Schulman et al. 2017]

PPO-PBRS – Per-Test Error



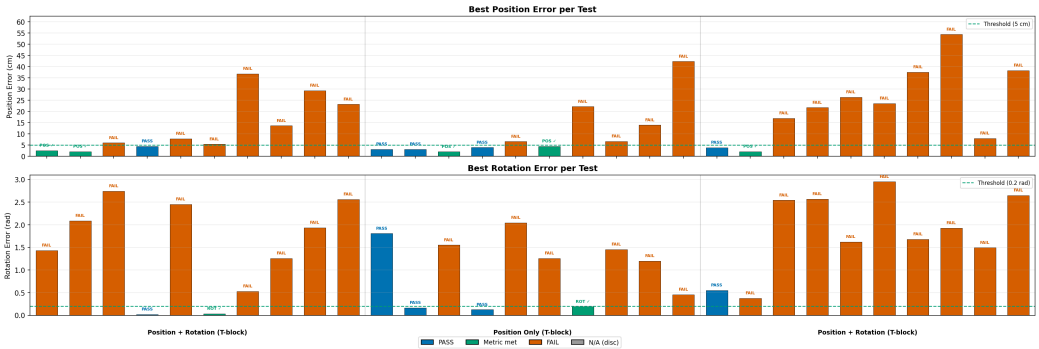
24/30 (80.0%) – 100% pos-only, 70% pos+rot. All 6 failures are rotation-limited (large-yaw / spin scenes); 8/10 rotation-dominant scenes solved.

PPO-Curriculum – Per-Test Error



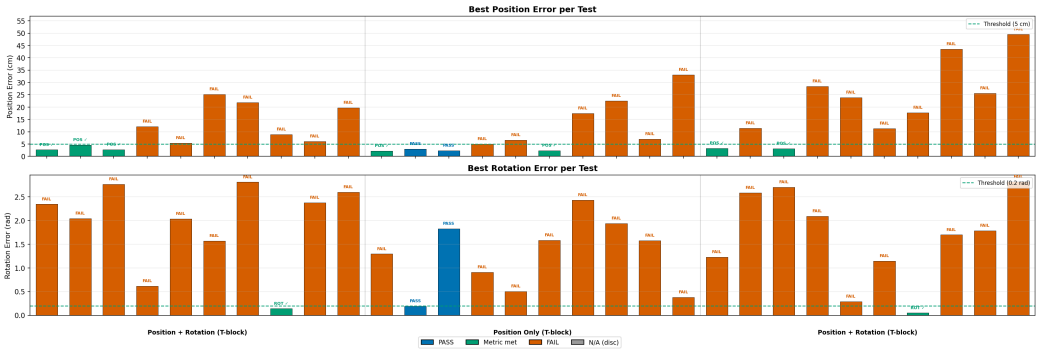
23/30 (76.7%) – same 100% pos-only, 65% pos+rot. Tighter position precision from Phase 1 (0.023 vs 0.032 m), but marginally worse rotation (0.663 vs 0.568 rad).

TASP-dPose – Per-Test Error



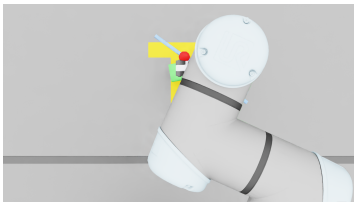
5/30 (16.7%) – 3 pos-only, 2 pos+rot. Only the easiest scenes pass, each with 1–2 successful trials; rotation never learned (RotErr 1.457 rad).

ASP-dPose – Per-Test Error

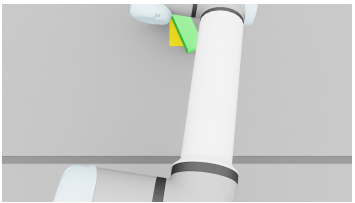


2/30 (6.7%) – only the two easiest position-only scenes pass. Zero pos+rot success; rotation never learned (RotErr 1.612 rad). Outcome-based Alice collapses entirely.

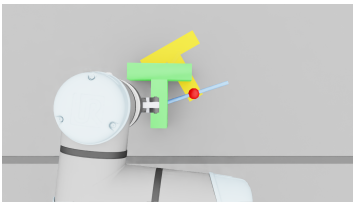
PPO-PBRS – Validation Runs (overhead)



Forward (pos-only) ▶ play clip

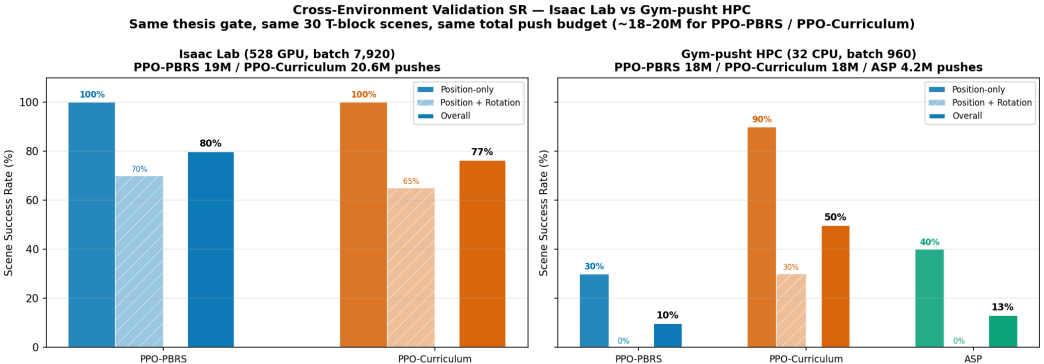


Lateral push (pos-only) ▶ play clip



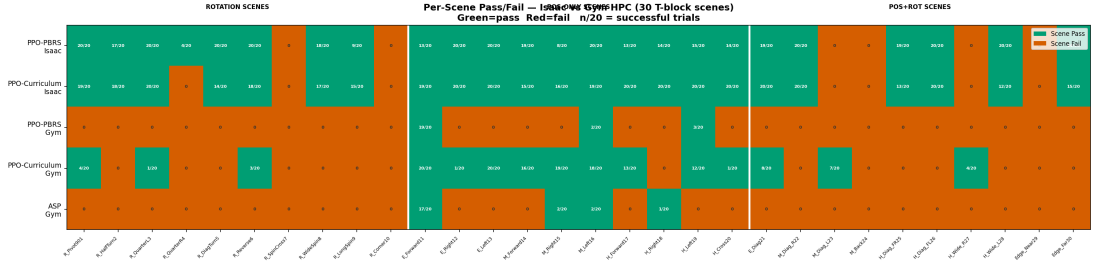
Translate + rotate ▶ play clip

Cross-Environment Validation – Isaac vs Gym



The ordering single-agent ≈ curriculum ≫ self-play holds in both simulators.

Per-Scene Outcomes – Isaac vs Gym



Green = scene solved (any of 20 trials), Red = failed

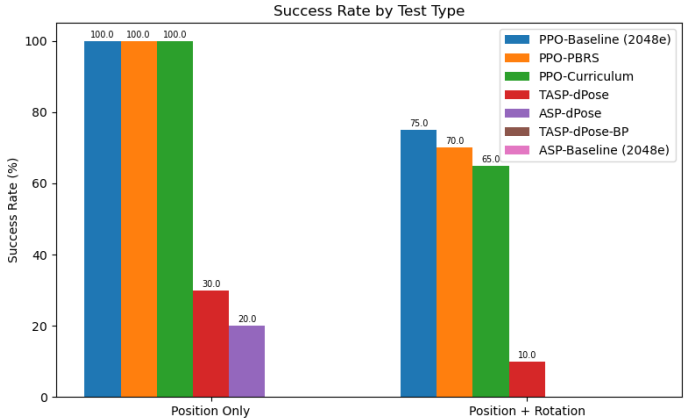
[Ng et al. 1999] [Grześ 2017] [Plappert et al. 2021]

7. Discussion

Discussion – What the Validation Suite Reveals

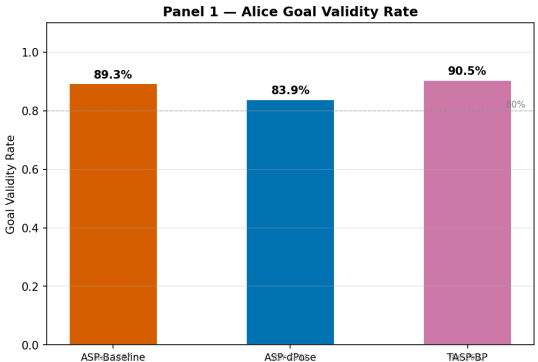
The combined pos+rot gate is the wall:

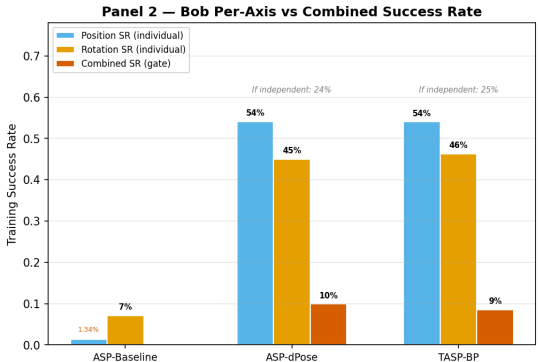
- Position-only scenes: solved by both single-agent models (100%).
- Pos+rot scenes: the bottleneck – PPO-PBRS 70%, self-play $\leq$10%.



[Mason 1986] [Goyal et al. 1991] [Ng et al. 1999]

ASP Diagnostic – Alice & Bob Training Dynamics





Left: Alice validity converges to 83–91% across all three ASP variants at different training budgets. **Right:** Bob learns position and rotation to ∼50% each under PBRS – but the combined gate caps at ∼10%, 2.5× below the independence product.

ASP – The Empirical Result

No ASP configuration exceeds 16.7% against an 80% single-agent ceiling:

ASP configuration	Valid SR	Pos+rot SR
Original fractional reward, 2,048 envs	0.0%	0%
PBRS outcome-based (ASP-dPose)	6.7%	0%
PBRS time-based (TASP-dPose)	16.7%	10%
PBRS time-based + Bob penalty	0.0%	0%
gym-pusht (different sim, same gate)	13.3%	0%
Single-agent ceiling (PPO-PBRS)	**80.0%**	**70%**

- Time-based Alice (TASP-dPose) is the best variant – 2.5× better than outcome-based, but still **4.8× below** single-agent
- Self-play and single-agent cost the same per iteration (∼0.022 it/s) – the gap is **structural**, not a budget artifact

Key Result – PBRS Substitutes for Curriculum

Model	Valid SR	vs Single-Agent
PPO-PBRS (single-agent)	**80.0%**	– (baseline)
PPO-Curriculum	76.7%	-3.3pp (n.s.)
TASP-dPose (time-based self-play)	16.7%	$4.8\times$ gap
ASP-dPose (outcome self-play)	6.7%	$11.9\times$ gap

The Substitution Finding (RQ3)

- **Curriculum adds no value at this scale:** PPO-Curriculum 76.7% vs PPO-PBRS 80.0% – 3.3pp gap **not significant** ($p > 0.05$). PBRS *substitutes* for the staged curriculum.
- **Regime-dependent:** holds at Isaac batch 7,920; at gym batch 960, curriculum *does* help (50% vs 10%).
- **Self-play (4.8–11.9$\times$ worse) helps in neither regime.** PBRS is the dominant lever – invest in reward design first.

8. Conclusion & Takeaways

Conclusion – Reward & Self-Play (RQ1–2)

RQ1 – PBRS enables efficient on-policy pushing

PBRS achieves 80.0% SR at 528 parallel envs – 4× fewer than the original fractional reward requires (2,048 envs) for the same performance. Dense, correctly-signed, bounded rewards on every push. Policy invariance theorem holds with stochastic PhysX.

RQ2 – Self-play fails structurally on this task

Five ASP configurations: **no variant exceeds 16.7%**. Single-agent ceiling: 80.0%. Self-play and single-agent cost the same per iteration – the gap is structural, not a budget artifact. Time-based Alice partially mitigates (+10pp) but cannot close the 4.8× gap.

[Ng et al. 1999] [Grześ 2017] [Plappert et al. 2021] **[Sukhbaatar et al. 2018]**

Conclusion – Curriculum Substitution (RQ3)

RQ3 – PBRS substitutes for curriculum at sufficient batch size

PPO-PBRS (80.0%) vs PPO-Curriculum (76.7%): 3.3pp gap is **not statistically significant** ($p > 0.05$). Substitution is **regime-dependent**:

- Isaac (batch 7,920): PBRS alone suffices – curriculum adds no value
- Gym (batch 960): curriculum helps (50% vs 10%) – PBRS gradient degraded by small-batch variance

[Devlin & Kudenko 2012] [Grześ 2017] [Narvekar et al. 2020]

Computational Overhead – Self-Play Is Not Slower

Controlled measurement (same machine, same 528 envs, same cuRobo IK + physics):
PPO-PBRS (single-agent) vs the self-play variants run at $\sim$**1.0**$\times$ per-iteration wall-clock ($\sim$0.022 it/s for both). The 2-agent machinery (Alice + Bob, GoalEncoder, ABC, historical pool) adds **no** measurable per-iteration overhead – the shared cuRobo IK + PhysX contact step dominates the cost.

Key result:

- Self-play is **not** slower per iteration – it simply **does not learn**
- The real cost of self-play is model/code complexity and added failure modes (two networks, non-stationary objective, ABC), *not* throughput
- So the 4.8$\times$ performance gap is **not** a compute-budget artifact – both used the same compute

[Plappert et al. 2021] [Schulman et al. 2017]

Extra Takeaway – ManagerBasedRL vs DirectRL

An engineering observation, separate from the scientific claims.

- Isaac Lab's **ManagerBasedRLEnv** (Action/Observation/Event/Termination managers) adds per-step Python dispatch; **DirectRLEnv** bypasses it with direct tensor ops.
- This is an *architecture* cost, independent of ASP – self-play and single-agent run at the same per-iteration time ($\sim 1.0\times$).

API	env-steps/s	relative
DirectRLEnv	*[to measure]*	*[run pending]*
ManagerBasedRLEnv	*[to measure]*	*[run pending]*

Controlled same-task measurement to be run; numbers will replace the placeholders. No multiplier is claimed until then.

[Makoviychuk et al. 2021] [Rudin et al. 2022]

Limitations & Future Work

1. **Bob penalty collapses to 0%** – penalty on catastrophe phase ends creates a "fail-fast" equilibrium.
   → **Zero-sum game theory** (Letcher 2019) may stabilise symmetric self-play; test penalty only on non-catastrophe ends.
2. **Combined pos+rot gate is the universal bottleneck** – joint gate rarely fires under self-play (0–10% pos+rot).
   → **Relaxed success gate** or partial credit per objective to provide gradient toward both.
3. **Single-seed validation** – training SR variance across 5 chains (BestSR 0.081–0.110).
   → **Re-evaluate** checkpoints across 5 seeds for validation confidence intervals.
4. **RQ3 substitution is regime-dependent** – PBRS substitutes at batch 7,920; curriculum helps at batch 960.
   → **Batch-size sweep** to map the substitution boundary and gradient-variance threshold.

[Letcher et al. 2019] [Dennis et al. 2020] [Schulman et al. 2017]

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Thank You – Questions?

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June 2026

Appendix Outline

- A. Push Mechanics – Limit Surface & Characteristic Length L
- B. Training curves for all PBRS models (Success Rate, Reward, Loss)
- C. PBRS parameter sensitivity (k_p, k_r, w_{pos}, w_{rot})
- D. Validation scene descriptions (30 test cases)
- E. cuRobo IK pipeline and workspace configuration
- F. Observation space specifications (all model variants)
- G. Per-test error bars – PPO-PBRS vs PPO-Curriculum head-to-head and all-model comparison (95% binomial CIs)
- **H. P82 Curriculum Trigger Bug** – why the original trigger never fired
- **I. BobPenalty Fail-Fast Equilibrium** – degeneracy mechanism
- **J. Confounded Ablation Disaggregation** – 7-axis variation table
- **K. Training Budget Comparison** – envs, iterations, push-macros
- **L. Validation Protocol** – scene set, trials, disclosed limitations

PBRS – The Shaping Theorem

Ng et al.:

$$\mathcal{R}'(s,a,s') = \mathcal{R}(s,a,s') + \underbrace{\gamma\Phi(s') - \Phi(s)}_{\text{shaping reward } F}$$

Policy invariance: $Q'(s,a) = Q(s,a) - \Phi(s)$ – optimal actions unchanged.

Our potential functions (PPO-PBRS / PPO-Curriculum):

$$\Phi(s) = w_{\text{pos}} e^{-k_p d_{\text{pos}}^2} + w_{\text{rot}} e^{-k_r c(s)}, \quad c(s) = \frac{1 - \cos(\theta^* - \theta)}{2}$$

$k_p{=}30,\ k_r{=}5,\ w_{\text{pos}}{=}w_{\text{rot}}{=}10.0$

Dense reward: $F(s,s') = \Phi(s') - \Phi(s)$ – difference only, no constant drain.

Episodic PBRS : $\gamma_{\text{shaping}}{=}1.0$ valid – $\sum F = \Phi(s_T) - \Phi(s_0)$ bounded.

[Ng, Harada & Russell 1999] **[Grześ 2017]** [Devlin & Kudenko 2012]

PBRS – Why It's the Right Choice

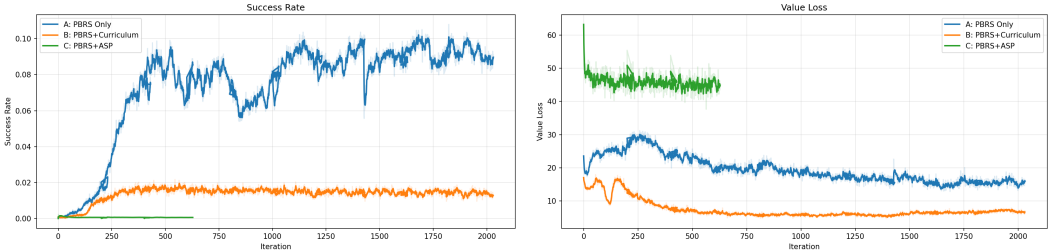
Properties of $F(s, s') = \Phi(s') - \Phi(s)$:

1. **Markovian** – depends only on s, s', not history.
2. **Bounded** – exponential potentials prevent reward explosions from PhysX contact spikes.
3. **Zero-sum cycles** – detour pushes net to zero; agent free to explore without reward hacking.
4. **No constant drain** – $F{=}0$ when stationary; no penalty for hesitation or deliberation.

Key consequence: you can replace any ad-hoc reward with a potential-based one *without changing what the agent learns* – only *how fast* it learns. Reward design and policy optimisation are decoupled.

[Ng, Harada & Russell 1999] **[Grześ 2017]** [Harutyunyan et al. 2015]

PPO-PBRS – Training Dynamics



Left: PPO-PBRS (blue) steadily improves training success rate over 3000 iterations. PPO-Curriculum (orange) follows a similar trajectory but converges slightly lower. Self-play variants remain near baseline throughout. **Right:** Stable value function – no explosion. Initial $V \approx 0.057$ with gain$= 0.01$.

[Schulman et al. 2017] [Ng et al. 1999]

PPO-Curriculum vs PPO-PBRS

Metric	PPO-PBRS	PPO-Curriculum	Gap
Validation SR	**80.0%**	76.7%	1.04×
PosErr (m)	0.032	**0.023**	B 28% lower
RotErr (rad)	0.568	0.663	A 14% lower
Pos-only SR	100%	100%	equal
Pos+rot SR	70%	65%	5pp

Result: Curriculum improves position precision (0.023 vs 0.032 m) from the pos-only phase, but slightly degrades rotation (0.663 vs 0.568 rad). Net effect: B is 3.3pp behind A. **Statistical note:** The 3.3pp A vs B gap is **not** significant ($p > 0.05$, two-tailed z-test; SE of difference = ±10.6pp across 30 independent scenes). Both A and B solve the task similarly well. **Takeaway:** A staged curriculum *can* work alongside PBRS, but does not beat the simpler no-curriculum baseline.

[Narvekar et al. 2020] [Portelas et al. 2020] [Florensa et al. 2017]

Self-Play – Validation Comparison

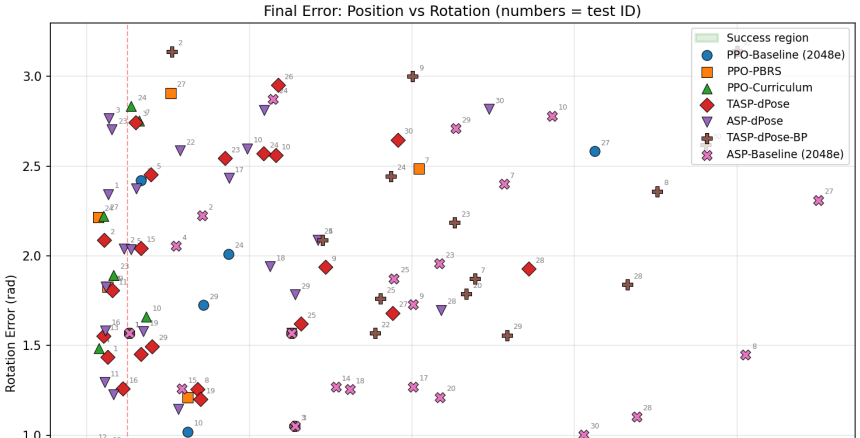
Model	Valid SR	PosErr	RotErr
PPO-PBRS (single-agent, ref.)	**80.0%**	0.032 m	0.568 rad
TASP-dPose (time-based)	**16.7%**	0.158 m	1.457 rad
ASP-dPose (outcome-based)	6.7%	0.143 m	1.612 rad

Key findings:

- Time-based self-play (TASP-dPose) is 2.5× better than outcome-based (ASP-dPose), but still **4.8× below** single-agent PPO-PBRS
- No self-play variant achieves pos+rot SR above 10% – the combined gate remains the bottleneck
- Self-play complexity provides no benefit

[Plappert et al. 2021] **[Sukhbaatar et al. 2018]** [Dennis et al. 2020]

Comparison Summary – Per-Scene Position × Rotation Error

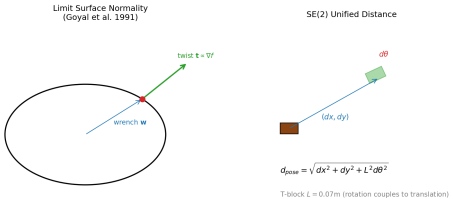


Appendix A – Push Mechanics: Limit Surface & L

Limit surface normality: $\mathbf{t} \propto \nabla f(\mathbf{w})$ – twist is normal to the surface, so every push couples translation *and* rotation.

Characteristic length L (radius of gyration):

- **T-block** $L = 0.07\,\text{m}$ – an off-centre push couples translation *and* rotation
- L is a physical constant, *not* a hyperparameter

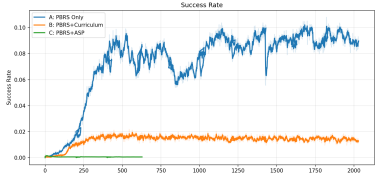


The T-block rotates under an off-centre push (coupled SE(2) motion)

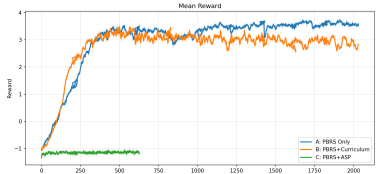
[Goyal et al. 1991] [Howe & Cutkosky 1996] [Mason 1986]

Appendix B – Training Curves (All PBRS Models)

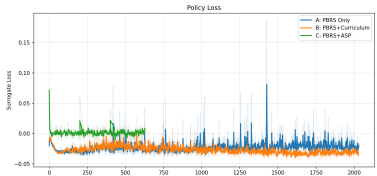
Success Rate



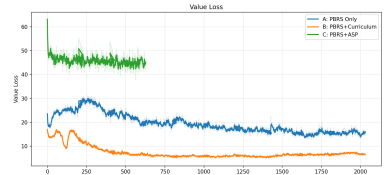
Mean Reward



Policy Loss



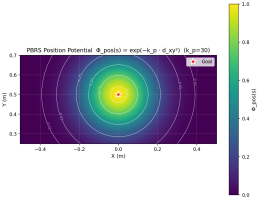
Value Loss



Appendix C – PBRS Theory Analysis

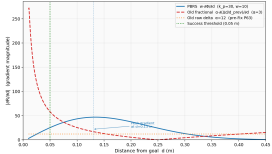
Potential Landscape

$$\Phi(s) = w \cdot \exp(-k \cdot d^2)$$

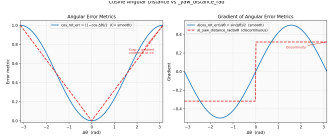


Gradient Comparison

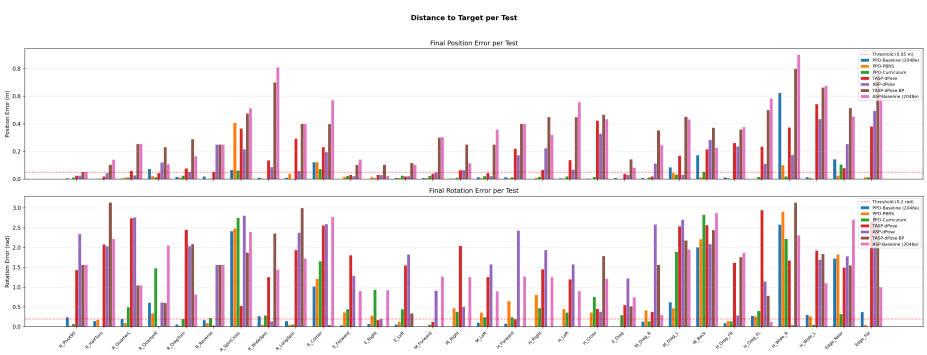
PBRS vs Fractional Formula



Cosine Angular Distance $c(s) = (1 - \cos(\theta^* - \theta))/2$



Appendix G – Per-Test Error Bars (95% Binomial CIs)



Top: PPO-PBRS (blue) vs PPO-Curriculum (orange) head-to-head – per-scene trial-level SR from 20 trials. Overlapping CIs on most scenes.

Bottom: All 4 models – the self-play variants are near-zero on all but the easiest scenes.

Green labels: position-only tests. **Purple labels:** position+rotation tests.

Appendix H – P82 Curriculum Trigger Bug

The original PPO-Curriculum trigger never activated because it gated on an unreachable metric.

Broken trigger (original):

If $\mathrm{mean}(\textit{per-push PosErr}) < 0.08$ m for ALL 50 consecutive iterations $\Rightarrow$ activate Phase 2

Why unreachable:

- Per-push PosErr is measured after *every* push, including early ones far from a freshly randomized goal (0.1–0.45 m away)
- The iteration-mean has a structural floor well above 0.08 m – even the *best* model's validation PosErr is 0.202 m
- 50-iteration AND gate: one noisy iteration resets the streak

Consequence: $w_{\text{rot}} = 0.0$ permanently – rotation received zero reward signal. Model B (the pre-P82 version) was effectively position-only PPO.
Fix (P82): gate on *episodic position-only SR* $\geq$ *0.5* over a 20-iteration lookback. This is a

Appendix I – BobPenalty Fail-Fast Equilibrium

Model I (TASP-dPose-BobPenalty) adds a Bob time penalty $R_B \mathrel{+}= -\gamma_{sp} \cdot t_B$ on *all* phase ends, including catastrophes.

Why it creates a degenerate equilibrium:

Trajectory	PBRS + Sparse	Penalty ($0.5 \times t_B$)	Total return
Fail on push 1	-10.5	-0.5	$\mathbf{-11.0}$
Fail on push 10	-15.0	-5.0	$\mathbf{-20.0}$
Succeed push 5	$+10.0$	-2.5	$\mathbf{+7.5}$

Mechanism:

1. PPO's gradient pushes Bob toward higher returns – "fail on push 1" (-11.0) beats "try for 10 pushes and fail" (-20.0)
2. Since failing is easier than succeeding in contact-rich physics, the policy collapses to immediate failure
3. Alice's reward collapses simultaneously: $R_A = 0.5 \cdot \max(0, 1 - 5) = 0$ – she gets zero signal

Appendix J – Confounded Ablation Disaggregation

Self-play variants differ from single-agent PPO-PBRS on $\geq$7 axes simultaneously:

Variable	PPO-PBRS (80.0%)	Self-play (6.7–16.7%)
Agent count	1	2 (Alice + Bob)
Goal distribution	Fixed random	Adversarial, shifting
GoalEncoder	None	ϕ-MLP 8D latent
ABC buffer	None	Yes (ASP-dPose) / No (TASP-dPose)
Historical pool	None	Yes
Episode budget	15 pushes/agent	5 (Alice) + 10 (Bob)
PPO updates/iter	1	2 (Alice, then Bob)

Implication: the 4.8$\times$ SR gap cannot be attributed to any single lever.

- TASP-dPose vs ASP-dPose isolates Alice's reward type – TASP-dPose gains +10pp
- But this is a single 1-axis experiment within a 7-axis package change
- We present "self-play as implemented," not "the curriculum mechanism in isolation"

[See thesis §6.3 for disaggregation analysis]

Appendix K – Training Budget Comparison

All PBRS variants used identical budgets. The PPO-Baseline used more compute for a direct architectural comparison.

Model	Envs	Max Iter	Est. Push-Macros
PPO-Baseline (Fractional)	2,048	100,000	∼3.07B
PPO-PBRS	528	3,000	∼23.8M
PPO-Curriculum	528	3,000	∼23.8M
ASP-dPose	528	3,000	∼23.8M
TASP-dPose	528	3,000	∼23.8M
TASP-dPose-BP	528	3,000	∼23.8M

Caveats:

- PPO-Baseline at 2,048 envs received ∼129× more push-macros than PBRS models
- Both converge to 80% validation SR – PBRS achieves this with 4× fewer parallel envs and a fraction of the total push budget

Appendix L – Validation Protocol

All models evaluated under an identical protocol:

- **Scene set:** 30 held-out T-block scenes – 10 rotation-dominant (R_*), 10 position-only (E_*/M_*/H_*), 10 combined position+rotation
- **Trials:** 20 independent trials per scene (best-of-20); a scene is "solved" if *any* trial meets the gate
- **Success gate:** pos_err < 0.05 m **and** rot_err < 0.2 rad at trial termination
- **Push budget:** up to 30 pushes per trial (training uses 5–15)
- **Scene SR vs Trial SR:** scene SR counts scenes where $\geq$1/20 trials pass; trial SR counts individual trial success rate

Disclosed limitations:

- **Single-seed validation:** each model evaluated once (one seed per checkpoint). Training SR variance exists across 5 chains (PPO-PBRS BestSR 0.081–0.110) but validation CIs are single-model.